Practical No. 7: Web Security with SSL/TLS

Aim

To **configure and implement secure web communication using SSL/TLS protocols**, including certificate management and secure session establishment.

Objectives

- To understand the concept of **Web Security**.
- To study **SSL (Secure Sockets Layer)** and **TLS (Transport Layer Security)** protocols.
- To understand the role of **SSL/TLS certificates**.
- To understand how encryption, authentication, and data integrity are achieved using TLS.

Theory

1. Web Security

- **Web Security** refers to the techniques and mechanisms used to protect websites, web applications, users, and data from unauthorized access and attacks.
- When users access websites over the Internet, sensitive information such as usernames, passwords, payment details, and personal information may be transmitted.
- Without encryption, attackers may be able to intercept and read this information.
- **SSL/TLS** is one of the fundamental technologies used to protect web communication.

2. SSL and TLS

SSL stands for **Secure Sockets Layer**, while **TLS** stands for **Transport Layer Security**.

- SSL/TLS is a cryptographic protocol used to provide secure communication between a client and a server.
- TLS is the modern successor to SSL.
- It protects data transmitted over networks such as the Internet.
- Websites using HTTPS generally use TLS to secure communication.

Example

When a user visits:

https://www.example.com

the browser and web server establish a secure TLS connection before exchanging protected application data.

3. Security Services Provided by SSL/TLS

SSL/TLS provides three major security services:

A. Confidentiality

- TLS encrypts data transmitted between the client and server.
- An attacker intercepting the communication cannot easily understand the encrypted data.

B. Authentication

- TLS certificates help authenticate the identity of the server.
- The browser can verify that the certificate was issued for the intended website and by a trusted Certificate Authority (CA).

C. Integrity

- TLS protects data against unauthorized modification during transmission.
- The communicating parties can detect if protected data has been altered.

4. SSL/TLS Certificate

An **SSL/TLS certificate**, commonly called an **SSL certificate**, is a digital certificate used to authenticate a website and establish encrypted communication.

A certificate generally contains information such as:

- Website/domain name
- Certificate issuer
- Public key
- Validity period
- Digital signature
- Certificate information

Certificate Authority (CA)

- A **Certificate Authority (CA)** is a trusted organization that issues and digitally signs certificates.

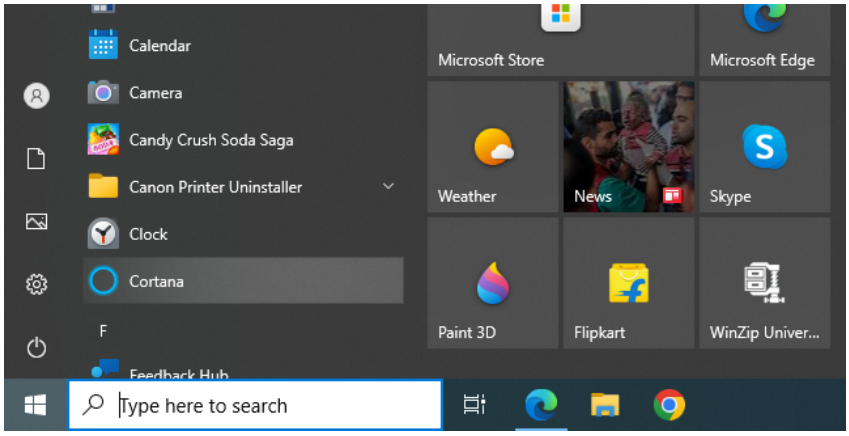
Examples include:

- DigiCert
- Let's Encrypt
- Sectigo

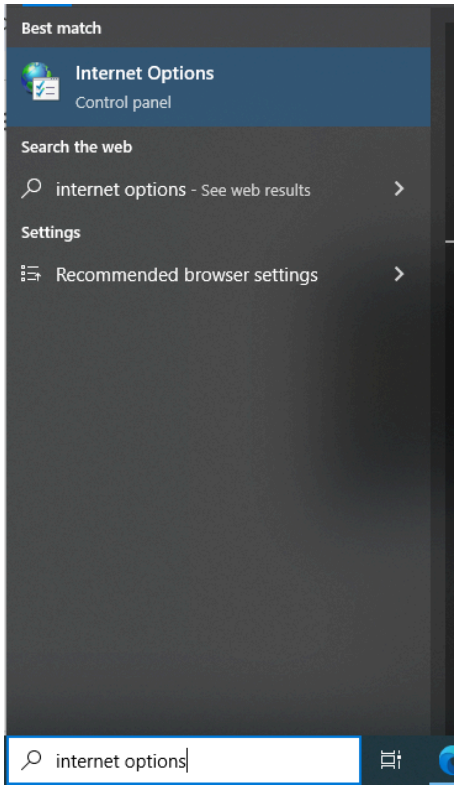
The browser maintains a list of trusted certificate authorities and uses it to help verify certificates.

Procedure: Configure SSL/TLS in Windows

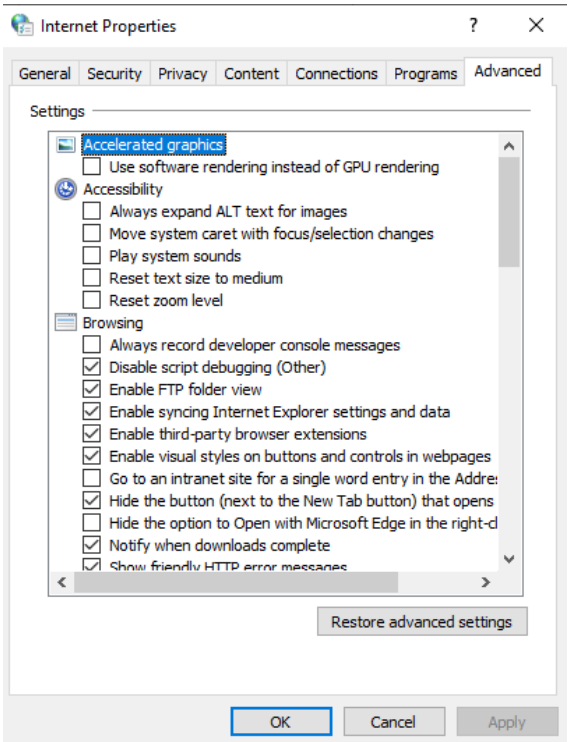
1. Open windows menu.



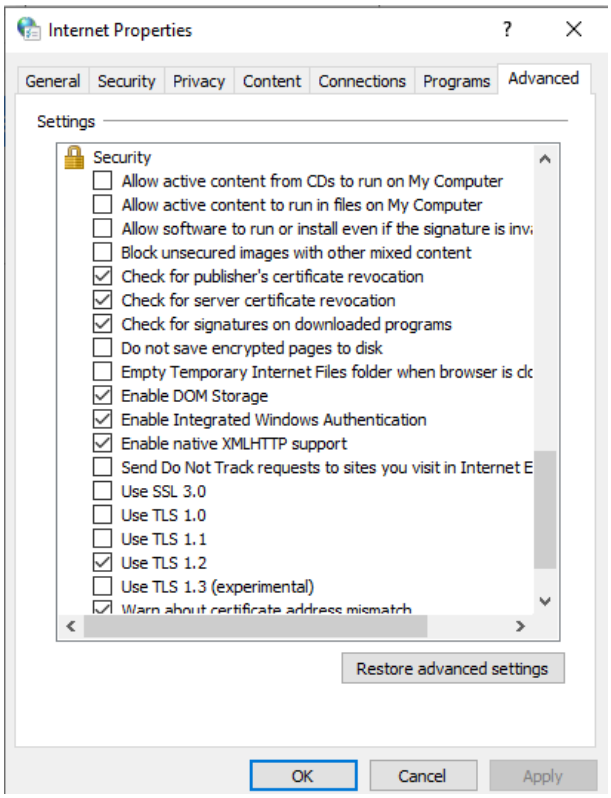
2. Search **Internet options** in the search bar and open it.



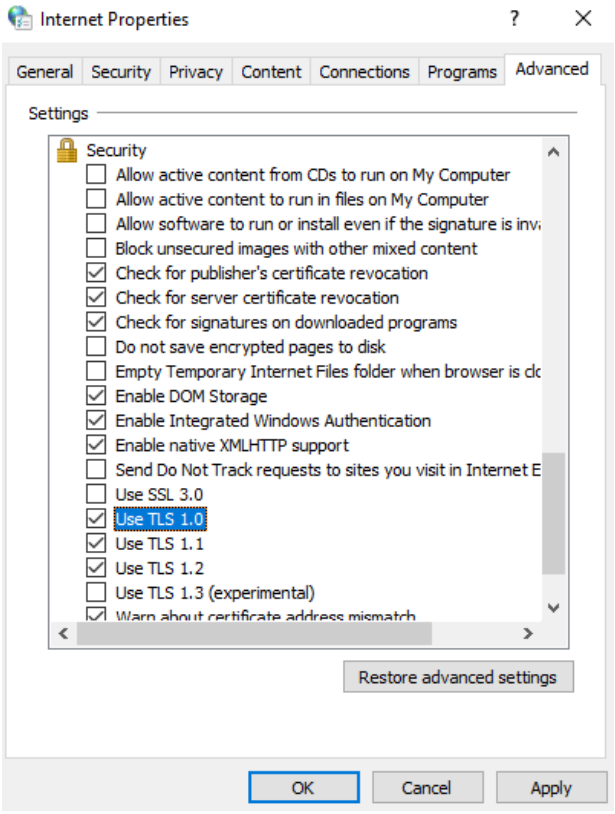
3. Now go to the **Advanced** tab.



4. Scroll down to the **Security** category.



5. Now **check the boxes** for your **SSL/TLS** version.



6. Click **OK.**
7. Restart the browser / PC.

Verification of Secure Communication

1. Open a modern web browser.
2. Visit an HTTPS website, for example:

https://www.example.com

3. Click the **site information/security icon** near the website address.
4. Verify that the connection uses **HTTPS**.
5. View the website's certificate information.
6. Check the certificate's:
    - Domain name
    - Issuer
    - Validity
    - Security status